

DATASET LOADING FROM KAGGLE

```
!pip install opendatasets
```

```
Requirement already satisfied: opendatasets in /usr/local/lib/python3.12/dist-packages (0.1.22)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from opendatasets) (4.67.1)
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (from opendatasets) (1.7.4.5)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from opendatasets) (8.3.1)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (6.3.0)
Requirement already satisfied: certifi>=14.05.14 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (2024.7.4)
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (3.4.0)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (3.11)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (5.29.5)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (2.9.0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (2.32.4)
Requirement already satisfied: setuptools>=21.0.0 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (75.1.0)
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (1.17.0)
Requirement already satisfied: text-unidecode in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (1.3)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (2.5.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (0.5.1)
```

```
import opendatasets as od
od.download("https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling/data")
```

Skipping, found downloaded files in "./churn-modelling" (use force=True to force download)

```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv("/content/churn-modelling/Churn_Modelling.csv")
df
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	
...	...	...	...	...	...	...	...	...	...	...	...	...
9995	9996	15606229	Obijaku	771	France	Male	39	5	0.00	2	1	
9996	9997	15569892	Johnstone	516	France	Male	35	10	57369.61	1	1	
9997	9998	15584532	Liu	709	France	Female	36	7	0.00	1	0	
9998	9999	15682355	Sabbatini	772	Germany	Male	42	3	75075.31	2	1	
9999	10000	15628319	Walker	792	France	Female	28	4	130142.79	1	1	

10000 rows × 14 columns

```
df1 = df.drop(columns = ['RowNumber', 'CustomerId', 'Surname',])
df1
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exit
0	619	France	Female	42	2	0.00	1	1	1	101348.88	
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	
3	699	France	Female	39	1	0.00	2	0	0	93826.63	
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	
...	...	...	...	...	...	...	...	...	...	...	...
9995	771	France	Male	39	5	0.00	2	1	0	96270.64	
9996	516	France	Male	35	10	57369.61	1	1	1	101699.77	
9997	709	France	Female	36	7	0.00	1	0	1	42085.58	
9998	772	Germany	Male	42	3	75075.31	2	1	0	92888.52	
9999	792	France	Female	28	4	130142.79	1	1	0	38190.78	

10000 rows × 11 columns

EDA

!pip install ydata-profiling

```
Requirement already satisfied: ydata-profiling in /usr/local/lib/python3.12/dist-packages (4.18.1)
Requirement already satisfied: scipy<1.17,>=1.8 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.16.3)
Requirement already satisfied: pandas!=1.4.0,<3.0,>1.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.2.3)
Requirement already satisfied: matplotlib<3.10,>=3.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.10.0)
Requirement already satisfied: pydantic<3,>=2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.12.3)
Requirement already satisfied: PyYAML<6.1,>=6.0.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (6.0.3)
Requirement already satisfied: Jinja2<3.2,>=3.1.6 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.1.6)
Requirement already satisfied: Visions<0.8.2,>=0.7.5 in /usr/local/lib/python3.12/dist-packages (from visions[type_image_patch]>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
Requirement already satisfied: minify-html>=0.15.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.18.1)
Requirement already satisfied: filetype>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.2.0)
Requirement already satisfied: phik<0.13,>=0.12.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.12.5)
Requirement already satisfied: requests<3,>=2.32.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.32.0)
Requirement already satisfied: tqdm<5,>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.67.1)
Requirement already satisfied: seaborn<0.14,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.13.2)
Requirement already satisfied: multimethod<2,>=1.4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.12)
Requirement already satisfied: statsmodels<1,>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.14.2)
Requirement already satisfied: typeguard<5,>=4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.4.4)
Requirement already satisfied: imagehash==4.3.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.3.2)
Requirement already satisfied: wordcloud>=1.9.4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.6)
Requirement already satisfied: dacite<2,>=1.9 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.2)
Requirement already satisfied: numba<0.63,>=0.60 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.60.0)
Requirement already satisfied: PyWavelets in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.2->ydata-profiling) (1.6.0)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.2->ydata-profiling) (10.4.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2<3.2,>=3.1.6->ydata-profiling) (3.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (1.3.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (4.54.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (25.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib<3.10,>=3.5->ydata-profiling) (2.9.0)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba<0.63,>=0.60 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.60.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2024.2)
Requirement already satisfied: joblib>=0.14.1 in /usr/local/lib/python3.12/dist-packages (from phik<0.13,>=0.12.5->ydata-profiling) (1.4.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (4.14.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=2->ydata-profiling) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (2.31.2)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.32.0->ydata-profiling) (2025.1.31)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels<1,>=0.13.2->ydata-profiling) (0.5.6)
Requirement already satisfied: attrs>=19.3.0 in /usr/local/lib/python3.12/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_image_patch]>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
Requirement already satisfied: networkx>=2.4 in /usr/local/lib/python3.12/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_image_patch]>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
Requirement already satisfied: puremagic in /usr/local/lib/python3.12/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_image_patch]>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib<3.10,>=3.5->ydata-profiling) (1.17.0)
```

```
from ydata_profiling import ProfileReport
prof = ProfileReport(df1)
prof.to_file(output_file='hitesh.html')
```

Summarize dataset: 100%45/45 [00:05<00:00, 7.91it/s, Completed]

0%| | 0/11 [00:00<?, ?it/s]

18%| | 2/11 [00:00<00:00, 18.43it/s]

64%| | 7/11 [00:00<00:00, 34.20it/s]

100%| | 11/11 [00:00<00:00, 33.10it/s]

Generate report structure: 100%1/1 [00:05<00:00, 5.46s/it]

Render HTML: 100%1/1 [00:00<00:00, 2.23it/s]

Export report to file: 100%1/1 [00:00<00:00, 24.19it/s]

FEATURE ENGINEERING

df1.sample(5)

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exit
6764	570	France	Male	37	6	0.00	1	1	1	187758.50	
8250	850	France	Female	35	9	0.00	2	0	0	25329.48	
3775	528	Germany	Male	22	5	93547.23	2	0	1	961.57	
7383	737	Spain	Female	39	7	130051.66	2	0	0	55356.39	
9244	546	Germany	Female	25	3	132837.70	1	1	0	131647.31	

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import MinMaxScaler
from sklearn.feature_selection import SelectKBest,chi2
from sklearn.linear_model import LogisticRegression
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline , make_pipeline
```

X_train , X_test , y_train , y_test = train_test_split(df1.drop(columns = ['Exited'],axis =1),df1['Exited'], test_size = 0.

X_train

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
7681	641	France	Male	33	2	146193.60	2	1	1	55796.83
9031	541	France	Male	39	7	0.00	2	1	0	19823.02
3691	590	France	Female	76	5	160979.68	1	0	1	13848.58
202	516	Spain	Male	50	5	0.00	1	0	1	146145.93
5625	508	France	Female	60	7	143262.04	1	1	1	129562.74
...	...	...	...	...	...	...	...	...	...	...
9225	594	Germany	Female	32	4	120074.97	2	1	1	162961.79
4859	794	Spain	Female	22	4	114440.24	1	1	1	107753.07
3264	738	France	Male	35	5	161274.05	2	1	0	181429.87
9845	590	Spain	Female	38	9	0.00	2	1	1	148750.16
2732	623	Germany	Female	48	1	108076.33	1	1	0	118855.26

7000 rows × 10 columns

y_train

	Exited
7681	1
9031	0
3691	0
202	1
5625	0
...	...
9225	0
4859	0
3264	0
9845	0
2732	1

7000 rows × 1 columns

dtype: int64

```
#onehotencoding
tnf1 = ColumnTransformer([('one_hot_encode',OneHotEncoder(sparse_output =False , drop = 'first',handle_unknown='ignore')),[:
```

```
#feature scaling
tnf2 = ColumnTransformer([('feature_scaling', MinMaxScaler(),slice(0,10))])
```

```
#feature selection
tnf3 = SelectKBest(score_func=chi2,k=6)
```

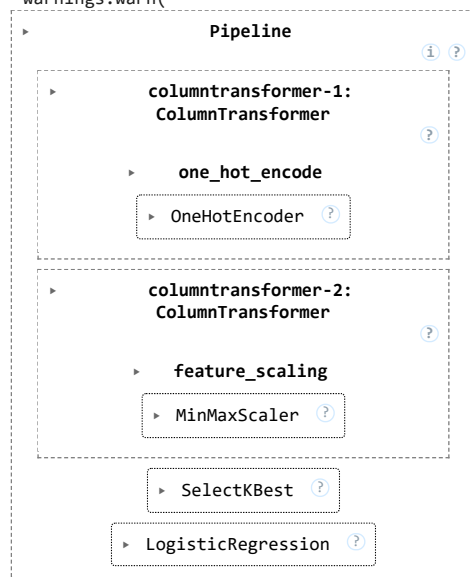
```
#train the model
tnf4 = LogisticRegression()
```

PIPELINE

```
pipe = make_pipeline(tnf1,tnf2,tnf3,tnf4)
```

```
pipe.fit(X_train,y_train)
```

/usr/local/lib/python3.12/dist-packages/sklearn/feature_selection/_univariate_selection.py:783: UserWarning: k=6 is greater than the number of features (5). This will result in a warning being raised during fit.



PREDICTION AND ACCURACY

```
y_pred = pipe.predict(X_test)
```

```
y_pred
```

```
array([0, 0, 0, ..., 0, 0, 0])
```

```
from sklearn.metrics import accuracy_score  
accuracy_score(y_test,y_pred)
```

```
0.793
```

EXPORTING THE PIPELINE

```
import pickle
```

```
pickle.dump(pipe,open('pipe.pkl','wb'))
```

TESTING

```
import pickle  
  
try:  
    with open('pipe.pkl', 'rb') as f:  
        loaded_pipe = pickle.load(f)  
        print("Pipeline loaded successfully!")  
        # You can now use 'loaded_pipe'  
        # For example, to check its type or content:  
        # print(type(loaded_pipe))
```